

W1-Lab

This lab consists of two parts:

- a. **Part 1:**
Review the video lessons and study the PowerPoint slides provided for this week. This material introduces key concepts that will be used throughout the course.
- b. **Part 2:**
Watch the virtual machine setup video and complete the steps to configure your Windows 11 VM. This setup is required to ensure you are prepared for upcoming hands-on labs.

Submission Requirements

Please submit one document that includes:

- Answers to all lab questions
- Required screenshots (snips) showing completion of the virtual machine setup tasks

Both parts must be completed and submitted to receive credit for this lab.

Part1:

Jimmy works at ***The Great Company***. One of the IT workers at this company has flagged a Windows workstation in the Accounting Department for suspicious activity:

1. Notepad (notepad.exe) was observed making an outbound network connection
2. Two unexpected processes were running on the system
3. Windows Security logs show multiple failed logon attempts followed by one successful logon.
4. The hard drive contains unusual files in non-standard locations.

No attempt has been made to analyze or fix the system.

Important Forensic Assumption

The forensic investigation is initiated the moment computer crime is suspected. From this point forward, all actions must follow forensically sound procedures. Considering the scenario explained above, please answer the following questions.

Questions

1. Why must a forensic investigation be initiated immediately once suspicious activity is detected?

Computer evidence can be fragile and disappear very quickly either due to automation by the system or by attackers. Starting the investigation immediately ensures that evidence is preserved properly

2. At what point does this situation transition from an IT issue to a computer forensic investigation?

It is transitioned from an IT issue to an investigation when a crime is thought to have occurred such as the outbound network connections or multiple failed login attempts.

3. What types of preliminary evidence should be identified and documented during the initial response?

System details, Dates and times of discovery, Active processes and services, Network connections, System/Security Logs, Suspicious Files and location of the files

4. Why is the goal of the immediate response documentation and preservation, not analysis?

Because preservation ensures that findings can be validated later as analysis can inadvertently change the system or destroy evidence

5. Under what circumstances would a court warrant for seizure be required in this scenario?

Device is owned personally and not by the company or LE is involved and the evidence will be used in a criminal investigation

6. Why must legal authorization be obtained before seizing or copying evidence, if required?

To ensure that evidence is obtained legally and admissible. Without doing so, an organization can risk violating privacy laws.

7. What first responder procedures should be followed when arriving at the suspected crime scene?

Ensure safety, secure the area, limit access to those who are only required, Identify those present, Photograph the system and workspace, Chain of Custody documents

8. Why is it important that first responders do not alter system state unnecessarily?

It can change timestamps, terminate any active malware processes, modify system logs, or damage other evidence integrity

9. What items in this scenario should be formally seized as evidence, and why?

Entire workstation or storage device, any media in the area such as USB drives or external hard drives, note pads or passwords in the area around the device as all of these things can be used to either prove or disprove malicious intent by the user. Such as if the removable media has malware on it, why did the employee have it?

10. Why must evidence be properly labeled, documented, and secured at the scene?

Maintain integrity and prove it was not tampered with. Proper labels also prevent any confusion and supports the chain of custody requirements.

11. Why must seized evidence be securely transported to a forensic laboratory?

Avoid any damages, tampering, or loss of data and ensures the evidence is maintained between seizure and analysis.

12. What risks exist if evidence is transported improperly?

Physical damage, loss or theft, malware automatically executing if the device is left powered on, tampering of evidence

13. Why should two bit-stream (forensic) copies of the evidence be created rather than a single copy?

Having a backup copy can be used for validation and recovery if the first copy (for analysis) is corrupted or damaged

14. What is the purpose of maintaining an original evidence copy that is never analyzed?

Allows for independent verification and protects against any claims of altering evidence

15. What information must be documented at each stage of this process to maintain evidence integrity?

Who handled the device, where the transfer was, condition of the device, actions performed and tools utilized for both imaging and analysis

16. What key elements should be included in the initial forensic report for this incident?

Case summary, timeline of events, system ID information, evidence list and chain of custody form, relevant findings, conclusion of findings

Key Takeaway

A forensic investigation is a process, not a reaction. Every step from suspicion to reporting must be deliberate, documented, and legally defensible.

Part2:

Please follow the posted video tutorial to set up both your Windows virtual machine and Kali Linux virtual machine.

Be sure to create your own personal user account on each system as instructed in the video.

To verify that your virtual machines are set up correctly, complete the following steps:

1. On your Windows virtual machine, open the Command Prompt, run the command

`whoami`

and submit a screenshot showing both the command and its output.

Important:

All work must be completed on **your own virtual machine**. Submitting screenshots or work from another person's system will be considered **plagiarism** and handled according to course policy.